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approaches allow the construction of composite indicators for different compensation degrees. On these premises, the aim of this paper is to illustrate the behaviour of some methodologies that build composite indicators allowing different compensation degrees. We analyse the results provided by each of these methods and which of them provide a more varied complementary information when considering both compensatory and non-compensatory scenarios. An illustrative example is used to visualise the results.

■ WA-03

Wednesday, 8:30-10:00 - Bulding A, Room 3A

Topics in Combinatorial Optimization III

Stream: Combinatorial Optimization
Invited session
Chair: *Heinrich Kuhn*

1 - A reinforcement learning-based operator selection in iterated local search for solving scheduling problems
Maryam Karimi Mamaghan, Patrick Meyer, Mehrdad Mohammadi, Bastien Pasdeloup

Machine learning techniques can be integrated into meta-heuristics for solving combinatorial optimization problems. The main aims of such integration are to improve the quality of the final solutions, increase the robustness of the meta-heuristics, and reduce the computational time. This work develops an iterated local search algorithm exploiting reinforcement learning to select search operators in the resolution of NP-hard scheduling problems. The candidate operators have different exploration/exploitation abilities, allowing the selection algorithm to adaptively select the most appropriate one during the search process, based on the status of the search and the performance history of each operator. Through comprehensive experiments, we demonstrate effective gains in using reinforcement learning rather than a random selection of operators. The results show that our algorithm provides better solutions in terms of optimality gaps and convergence behavior, without imposing significant computational overhead, compared to other meta-heuristics in the literature.

2 - A Hybrid Adaptive Large Neighborhood Search for Vehicle Routing Problems with Location Decisions
Stefan Voigt, Heinrich Kuhn, Markus Frank, Pirmin Fontaine

We present a novel heuristic for a class of Vehicle Routing Problems with Location Decisions based on the recently proposed Hybrid Adaptive Large Neighborhood Search (HALNS). We show that the Location Routing Problem (LRP) and the Multi-Depot Vehicle Routing Problem (MDVRP) can be formulated as special cases of the Two-Echelon Vehicle Routing Problem (2E-VRP) and therefore be readily solved with a single heuristic framework. The HALNS exploits the structure of such two-stage problems, where the first stage (location decision) strongly influences the second stage (routing decision). Once the first-stage decisions are made, the two-stage structure makes it hard to escape local minima in the second stage without allowing for major deterioration of the solution. The HALNS uses a population of solutions generated by an efficient ALNS to overcome this issue. Individuals of this population are subject to a crossover and selection phase, using elements of genetic algorithms resulting in a hybrid heuristic. Computational experiments on several sets of instances from literature show the good performance of the HALNS. The HALNS is on par with heuristics that are dedicated either to the LRP, the MDVRP or the 2E-VRP, showing that generality does not necessarily decrease solution quality. The HALNS especially outperforms all existing pure ALNS implementations on these problem classes, demonstrating the value of hybridization.

■ WA-04

Wednesday, 8:30-10:00 - Bulding A, Room 3B

Routing applications - onsite

Stream: Routing, Logistics, Location and Freight Transportation
Invited session
Chair: *Farzaneh Rajabighamchi*
Chair: *Anne-Laurence Hulot*

1 - An Exact Model for the order picker routing problem in warehouses
Farzaneh Rajabighamchi, Stan Van Hoesel, Christof Defryn
This paper is dealing with the order picker routing problem within a multi-block warehouse layout with several aisles. In the literature, exact algorithms only exist for small warehouses with few cross aisles (typically two or three), while for larger warehouse configurations a series of heuristic and meta-heuristic methods are available. We present a novel pre-processing algorithm for graph reduction by eliminating the extra corner vertices and aisles from the graph of warehouse locations and aisles. This allows us to solve this routing problem with a general TSP solver while significantly reducing running times. The presented method allows us to solve adequately big (more realistic) instances exactly. The algorithm is implemented and evaluated experimentally on a set of problem instances from the literature. The computational results illustrate that the proposed model outperforms existing formulations in terms of simplicity, size, and calculation time. Our mathematical model gives an optimum solution for all the instances, while the network size could be reduced by almost 73% on average.

2 - An Infeasible Space Exploring Matheuristic for the Production Routing Problem
Eleftherios Manousakis, Grigoris Kasapidis, Chris Kiranoudis, Emmanouil Zachariadis
We propose a novel matheuristic algorithm for solving the Production Routing Problem (PRP). The PRP is a hard-to-solve combinatorial optimization problem with numerous practical applications in the field of freight transportation and logistics. A manufacturer is responsible for determining production decisions, as well as the timing and the quantity of replenishment services offered to a set of geographically dispersed customers over a multi-period time horizon. The problem calls for jointly optimizing the production, inventory, distribution and routing decisions. The paper provides a two-phase infeasible space matheuristic algorithm for solving the PRP. The first phase deals with a relaxation of the problem to construct production-distribution plans. In the second phase, these are completed with routing information and optimized via a local search framework which oscillates between the feasible and the infeasible solution space. The framework is equipped with mixed integer programming components for optimally restoring infeasibility and for diversifying the conducted search. Computational experiments demonstrate that the infeasibility space exploration significantly contributes to the quality of the final solutions. The results obtained by the proposed matheuristic out-match the results of state-of-the-art approaches: 594 and 55 new best solutions out of 1440 and 90 instances of two well-established benchmark data sets of small-medium and large instances, respectively.

3 - Dynamic Technician Routing Problem with Stochastic Requests
Dai Trong Pham, Gudrun Kiesmuller
Motivated by the problem faced by many home-attended maintenance service providers, this paper considers the problem of routing and spare-parts planning of a single service technician under a dynamic environment where stochastic information about future requests, such as location and demand for spare-parts, is revealed over time. We designed and tested our computationally-tractable simheuristic based solution approach. The preliminary results show that incorporating stochastic information yield better results, compared to the myopic approach where no stochastic information about future requests is included.